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EXPERIMENT 12.1.1 PHONE NUMBER DATABASE MANAGEMENT

ALGORITHM:

- **Start**
- **Input** an integer n (number of operations)
- **Initialize** an empty dictionary contacts
- **Repeat** steps 5–10 for i = 1 to n:
- **Input** the operation as a string
- **Split** the input into parts
- **Check operation type:**
- **If operation = "ADD":**

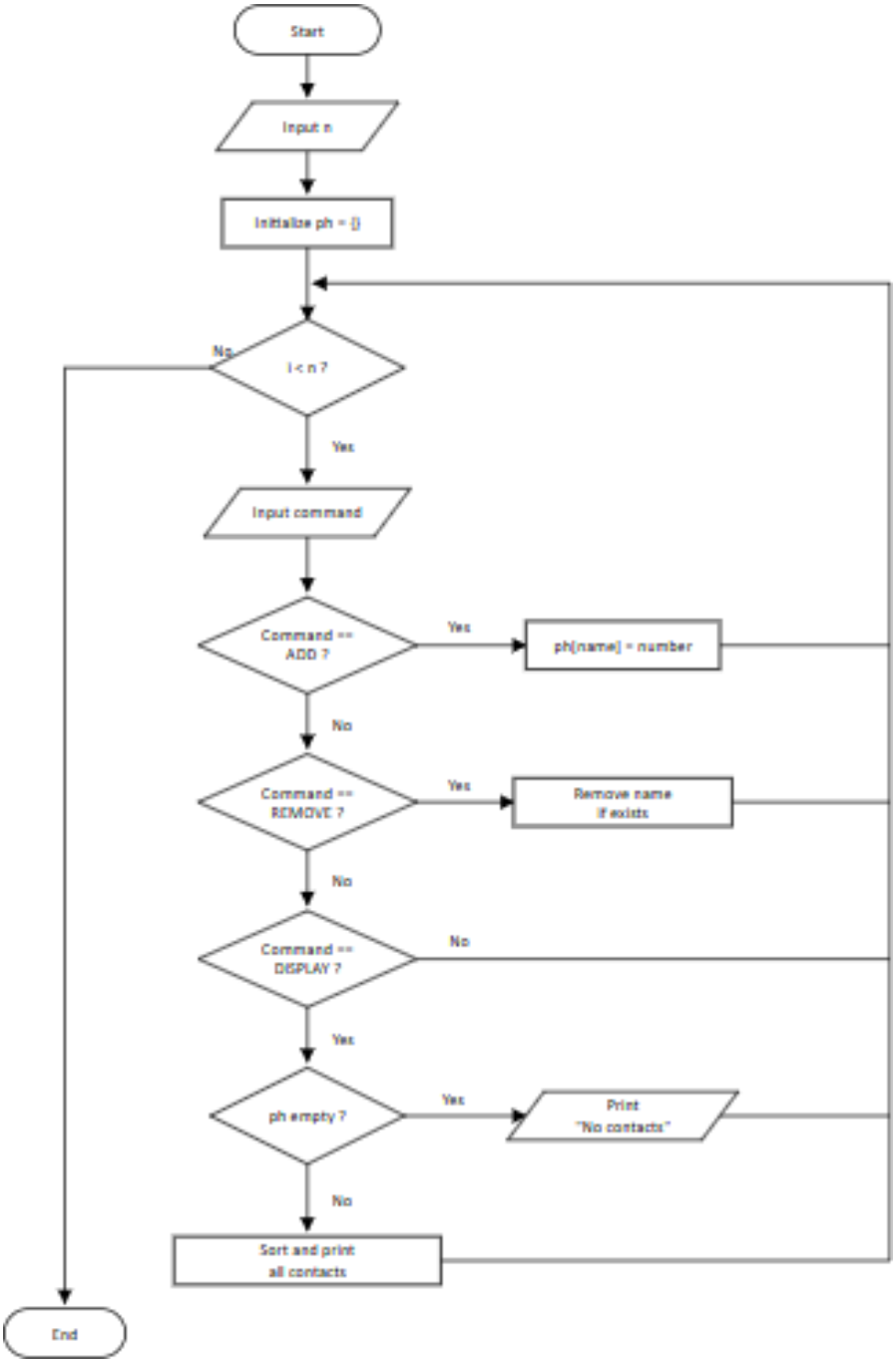
- Read name and phone_number
- Insert into dictionary: contacts[name] = phone_number
- (If name already exists, it will automatically update)

b) Else if operation = "REMOVE":

- Read name
- If name exists in dictionary:
  - Delete it using del contacts[name]
- Else:
  - Do nothing

c) Else if operation = "DISPLAY":

- If dictionary is empty:
  - Print "No contacts"
- Else:
  - Get all keys (names)
  - Sort names in alphabetical order
  - For each name:
    - Print <name>: <phone_number>
- **End of loop**
- **Stop**


Start
Input n
Initialize ph = {}
i < n ?
No
Yes
Input command
Command == ADD ?
Yes
ph[name] = number
No
Command == REMOVE ?
Yes
Remove name if exists
No
Command == DISPLAY ?
No
Yes
ph empty ?
Yes
Print "No contacts"
No
Sort and print all contacts
End

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12.1.1. Phone Number Database Management

02:55

Write a Python program to manage a phone number database using a dictionary. The dictionary stores contact names as keys and phone numbers as values.

The program should support three operations: adding a contact (if the name already exists, update the phone number), removing a contact (if the name does not exist, ignore the operation), and displaying all contacts in alphabetical order by name.

Input Format:

1. First line contains an integer n representing the number of operations.
2. Each of the next n lines contains a single operation in one of the following formats:
   - "ADD <name> <phone_number>" where <name> is a string, <phone_number> is a string of 10 digits.
   - "REMOVE <name>" - name is a string.
   - "DISPLAY" - no additional parameters.

Output Format:

- For display operation, print all contacts in alphabetical order by name in the format:

```
<name>: <phone_number>
```

- If the database is empty during display operation, print: "No contacts".
- Do not print anything for add and remove operations.

Constraints:

- Name contains only alphabetic characters (no spaces). And it is case sensitive.

Explorer

phoneDat...

Submit

```python
# Phone Number Database Management

n = int(input())
contacts = {}

for _ in range(n):
    command = input().split()

    if command[0] == "ADD":
        name = command[1]
        phone = command[2]
```

Average time 0.008 s 7.88 ms

Maximum time 0.011 s 11.00 ms

5 out of 5 shown test case(s) passed

3 out of 3 hidden test case(s) passed

Test case 1 11 ms Debug

Expected output	Actual output
4	4
ADD Alice 9876543210	ADD Alice 9876543210
ADD Bob 8765432109	ADD Bob 8765432109
ADD Charlie 7777777777	ADD Charlie 7777777777
DISPLAY	DISPLAY
Alice: 9876543210	Alice: 9876543210

Terminal Test cases